

The NFSC Four-Dimensional Pedagogy: A Progressive Teaching Model for Graduate Engineering Education

— A Case Study of "Solid Waste Pollution Prevention and Control Technology"

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Abstract: Graduate engineering education faces a persistent structural challenge: how to transform students from passive knowledge receivers into active problem solvers capable of making multidimensional engineering judgments. This paper presents the NFSC (Narrative - Framework - Scenario - Creation) four-dimensional pedagogy, a progressive teaching model designed to bridge this gap. The pedagogy employs documentary narratives to trigger cognitive engagement, analytical frameworks to equip students with diagnostic tools, immersive scenarios to temper application capabilities, and creative outputs to close the learning loop. Taking the graduate course "Solid Waste Pollution Prevention and Control Technology" at Chengdu University as a case study, this paper elaborates on the theoretical grounding, instructional design, and eight-week implementation of the NFSC model. Process data from three teaching cycles (2023 - 2025) provide qualitative evidence of cognitive transformation, including shifts in attribution patterns (from intuitive "blame the villagers"

to framework-informed analysis), longitudinal growth in multi-dimensional framework usage, and diversification of final outputs across research, design, and pedagogical reflection tracks. The NFSC model's honest documentation of limitations—small class size, absence of controlled comparison, single-course scope—positions it as a "fieldnote" contribution rather than a causal claim. The paper discusses transferability preconditions and boundaries for other engineering disciplines.

****Keywords:**** engineering education; graduate teaching; narrative pedagogy; scenario-based learning; constructive alignment; four-dimensional teaching model

1. Introduction

Graduate engineering courses bear the core mission of cultivating "problem solvers" rather than "knowledge receivers." Yet the dominant lecture-based paradigm—organizing instruction along the linear path of "definition → classification → principles → processes → cases"—proves insufficient for this goal [1]. Textbook cases are "clean problems": abstracted, idealized, and stripped of contextual complexity. Real-world solid waste management, by contrast, presents "wicked problems": a landfill siting decision involves not only geological and hydrological constraints but also community opposition, trans-jurisdictional accountability ambiguity, and temporal delays in environmental benefit realization.

This gap between pedagogical abstraction and engineering reality is not unique to solid waste courses. It reflects a broader structural tension in graduate engineering education: the need to train students for multidimensional decision-making under constraints, while the traditional lecture format excels primarily at efficient knowledge

transmission [2]. Engineering accreditation standards worldwide increasingly demand competency-based outcomes—what students can *do*, not merely what they *know* [3,4]. Yet translating abstract outcome statements (e.g., “analyze multidimensional mismatches in engineering problems”) into concrete, week-by-week instructional sequences remains an unresolved “last mile” problem [5].

This paper presents the NFSC (Narrative - Framework - Scenario - Creation) four-dimensional pedagogy, a teaching model developed and iterated over three years (2023 - 2025) in the graduate course “Solid Waste Pollution Prevention and Control Technology” at Chengdu University, China. The pedagogy draws upon documentary network novels published by the course instructor as a narrative resource pool, employs a domain-specific “Mismatch - Repair” analytical framework, deploys three types of immersive scenarios, and culminates in a three-track creative output system. The paper makes three contributions: (1) a theoretical derivation of the four-dimensional cognitive progression grounded in established learning theories; (2) a detailed account of the eight-week implementation with process-level evidence; and (3) an honest methodological reflection that positions this work as “fieldnote” scholarship rather than causal experimental research.

2. Theoretical Framework

2.1 The Four-Dimensional Cognitive Progression

The NFSC pedagogy structures the teaching - learning process into four progressive dimensions, each activating a distinct cognitive mode and building upon the previous one. The progression is not a simple temporal sequence but a qualitative transformation in how students engage with knowledge.

****Narrative (叙)**** - The first dimension triggers learning motivation and emotional resonance. Drawing upon Bruner’s narrative theory of

mind, which argues that humans organize experience primarily through narrative structures rather than logical propositions [6], NFSC replaces decontextualized case summaries with documentary narrative excerpts. Seven documentary network novels published by the course instructor, comprising approximately two million characters of non-fiction prose, provide 21 teaching excerpts (coded S001 – S021) that record authentic solid waste management scenarios across China: land devaluation around landfill sites, technology-centrism in environmental impact assessments, implementation failures in rural waste classification, and dignity claims in community NIMBY conflicts [7]. Students read “things that happened,” not “cases that were constructed.”

****Framework (框)**** - The second dimension transforms narrative-elicited affective engagement into structured analytical capability. Building on constructivist learning principles [8], NFSC introduces a domain-specific “Mismatch-Repair” framework comprising six sub-frameworks (spatial mismatch, siting justice, technological mismatch, waste time, behavioral mismatch, and humanistic mismatch, totaling 34 diagnostic dimensions) [9]. The framework serves as a cognitive scaffold: students first analyze cases intuitively, then re-analyze using the framework, and finally compare their pre- and post-framework analyses—experiencing what Perkins termed the “constructivist marriage” of intuitive engagement with structured tool use [10].

****Scenario (境)**** - The third dimension tempers analytical capability through immersive practice. Situated learning theory posits that knowledge acquisition cannot be divorced from its application context [11]. NFSC deploys three scenario types: (a) environmental impact assessment debates (Week 3), where students role-play as technical experts, regulatory reviewers, and community representatives; (b) Nudge-based behavioral intervention design (Weeks 5 – 6), where students design and simulate policy instruments; and (c) mock environmental court hearings (Week 7), where students experience the tension between procedural legality and substantive justice. Each scenario includes role cards, randomized event cards, and instructor

facilitation guides.

****Creation (創)**** - The fourth dimension closes the learning loop through creative production. Outcome-Based Education (OBE) principles require that all instructional activities ultimately serve demonstrable student capabilities [4,5]. NFSC offers three parallel output tracks: Track A (research paper)—students apply the five-dimensional unified diagnostic matrix to analyze a solid waste problem and discuss framework limitations; Track B (engineering design)—students complete a process design for a landfill/incineration/composting facility; Track C (pedagogical reflection)—students conduct meta-analysis of their own eight-week learning experience. All three tracks share common requirements: framework application, methodological notation, and honest discussion of framework gaps.

2.2 Progression Logic: From Reception to Production

The four dimensions form a cognitive ladder aligned with Bloom's revised taxonomy [12]. Narrative activates the "remember" and "understand" levels, but unlike traditional lectures where memorization targets abstract definitions, NFSC anchors understanding in emotionally charged narrative episodes. Framework elevates cognition to "apply" and "analyze" levels, transforming intuitive judgment into structured diagnosis. Scenario pushes toward "evaluate" by forcing students to make decisions under role constraints, time pressure, and incomplete information. Creation reaches "create"—students produce original work that extends beyond the taught material. This progression aligns with Laurillard's "conversational framework" for university teaching, where learning is conceived as iterative cycles of discursive engagement, adaptive practice, and productive reflection [13].

3. Method: The Eight-Week Implementation

3.1 Course Context

"Solid Waste Pollution Prevention and Control Technology" is a graduate elective at Chengdu University's Department of Environmental Engineering (course code 026039003, 2 credits, 32 contact hours—24 theoretical + 8 immersive practice). The course has undergone three iterative revisions since its 2023 launch: v1.0 (2023, traditional lecture), v2.0 (2024, pilot NFSC structure), and v3.0 (2025, full NFSC implementation). The current version serves approximately 30 graduate students per cycle, with heterogeneous prior training (approximately 60% from environmental engineering backgrounds, 40% from cross-disciplinary backgrounds including chemistry, materials science, and ecology).

3.2 Weekly Progression

****Week 1 (Narrative - Spatial Mismatch I):**** Students pre-read excerpts S001 - S003 from **Longqi Bay**, a documentary novel chronicling a landfill project's impact on surrounding communities. In-class discussion centers on one question: "What exactly did the villagers lose?" The instructor introduces the five-dimensional spatial mismatch framework (siting error, planning lag, capacity imbalance, transport paradox, boundary ambiguity) only after students have articulated their own observations.

****Week 2 (Framework - Spatial Mismatch II):**** Students re-analyze Week 1's excerpts using the siting justice four-dimensional framework (geological capacity, environmental capacity, social acceptability, intergenerational equity). The key pedagogical move is the "before - after comparison": students contrast their Week 1 intuitive analyses with their Week 2 framework-guided analyses.

****Weeks 3-4 (Scenario-Technological Mismatch):**** EIA debate scenario. Students role-play three stakeholder positions around an incinerator project's technology applicability. The three-type technological mismatch framework (over-design, under-application, directional error) is introduced post-debate as a diagnostic tool.

****Weeks 5-6 (Scenario-Behavioral Mismatch):**** Nudge intervention design. Using excerpts from **Willow Dam and Liu Family Bay**, students diagnose why a rural waste classification program failed despite adequate infrastructure. The four-factor behavioral mismatch framework (cognitive barrier, convenience trap, norm deficit, feedback delay) is introduced after students' intuitive diagnoses.

****Week 7 (Scenario-Humanistic Mismatch):**** Mock environmental court. Students litigate a case involving procedural versus substantive justice in a waste facility siting decision. The three-dimensional humanistic mismatch framework (distributive justice deficit, procedural justice deficit, recognition justice deficit) provides the analytical lens.

****Week 8 (Creation-Integration):**** Students apply the five-dimensional unified diagnostic matrix (Space $\times$ Technology $\times$ Behavior $\times$ Humanism $\times$ Time) to a comprehensive case, and submit their chosen track output.

3.3 Digital Infrastructure

A zero-configuration learning sub-portal ("Learning Garden") was developed alongside the course, providing weekly preview materials, a localStorage-based progress tracking system, four-section weekly learning pages (Preview $\rightarrow$ Learn $\rightarrow$ Assignment $\rightarrow$ AI Empowerment), and AI-assisted learning prompts [7]. The portal's static HTML/CSS/JS architecture requires no user registration, server-side logic, or database—students access it directly via browser.

4. Results: Fieldnote Evidence

The following evidence is presented as qualitative “fieldnote” documentation rather than causal experimental proof. The course served approximately 30 students per cycle across three cycles (2023–2025), with no parallel control group.

4.1 Cognitive Shift in Causal Attribution

During Week 5’s behavioral mismatch instruction, students were asked to intuitively diagnose why a rural waste classification program failed. Among approximately 30 students, “low villager quality” (素质低) was the most frequent attribution (11 students, ~37%). After the four-factor behavioral mismatch framework was introduced, students re-analyzed the same case. By Week 8, when a similar scenario was presented, only 2 students (~7%) invoked “low quality” attribution, while 7 students cited “convenience trap” and 6 cited “feedback delay”—demonstrating framework-informed analysis replacing intuitive moral judgment. This shift was not achieved through moral instruction but through the provision of analytical tools [14].

4.2 Longitudinal Growth in Framework Usage

Using the five-dimensional unified diagnostic matrix as a measurement tool, three time-point assessments were conducted. At Week 1 baseline, approximately 15 students could accurately use framework sub-dimension terminology. At Week 4 mid-point, approximately 25 students used at least two framework dimensions. At Week 8 end-point, approximately 12 students spontaneously employed three or more dimensions for cross-dimensional interaction analysis. Framework integration capacity emerged progressively across the eight-week cycle.

4.3 Output Diversification

Across three teaching cycles, Track A (research paper) accounted for approximately 40% of final outputs, Track B (engineering design) approximately 35%, and Track C (pedagogical reflection) approximately 25%. Notably, approximately 20% of students demonstrated emerging metacognitive awareness in their final reflections, articulating specific changes in their own analytical frameworks (e.g., "I used to think solid waste problems were purely technical; now I see they are simultaneously spatial, institutional, and human problems").

5. Discussion

5.1 Transferability Preconditions

The NFSC pedagogy's transferability to other engineering courses depends on three preconditions. First, the course must possess a "narrativizable" resource pool—not necessarily published documentary novels, but authentic, emotionally textured engineering case records. Second, the core knowledge domain must be abstractable into one or more analytical frameworks that serve as cognitive scaffolds. Third, instructors must be capable of designing and facilitating scenario-based teaching, which represents the highest-demand component of the NFSC model.

The "Mismatch-Repair" framework, while domain-specific to solid waste management, exemplifies a transferable design principle: using "mismatch" as an epistemological stance to reframe engineering problems from "technical failure" to "system-context misalignment." This principle is adaptable to hydraulic engineering (spatial-institutional water resource mismatches), transportation engineering (behavioral-demand mismatches in network planning), and beyond [15].

5.2 Honest Limitations

This study carries significant limitations that must be stated transparently. First, the evidence base consists of process data from a single course at a single institution across three teaching cycles, with no controlled experimental design. The documented cognitive shifts cannot be causally attributed to the NFSC pedagogy alone—novelty effects, instructor personality, and cohort-specific factors remain uncontrolled confounds. Second, the course scale (approximately 30 students per cycle) precludes statistical significance testing. Third, the documentary novel resource pool represents a high-barrier entry condition: the seven novels required over a decade of writing and publication, a commitment unlikely to be replicated by other instructors at comparable scale.

These limitations are not presented as disclaimers to be minimized but as constitutive features of a “fieldnote” approach to pedagogical scholarship. The value of this work lies not in causal inference but in honest documentation—recording a real classroom’s cognitive trajectory across three iterative cycles, with all its messiness, partial evidence, and unresolved questions intact.

6. Conclusion

The NFSC four-dimensional pedagogy offers a coherent, theoretically grounded, and practically implemented teaching model for graduate engineering education. Its core contribution is the articulation of a cognitive progression—from narrative-triggered engagement through framework-guided analysis and scenario-tempered application to creation-based closure—that addresses the structural gap between knowledge transmission and problem-solving capability in graduate engineering classrooms.

The model’s honest acknowledgment of its limitations—small scale,

absence of controls, resource-intensity barriers—positions it as an invitation rather than a prescription. Other engineering educators are encouraged to adapt the NFSC design principles (narrative anchoring, framework scaffolding, scenario immersion, creation closure) to their own disciplinary contexts, with the understanding that the specific implementation must be re-parameterized for each new domain.

References

- [1] ANDERSON T, DRON J. Three Generations of Distance Education Pedagogy[J]. The International Review of Research in Open and Distributed Learning, 2011, 12(3): 80–97.
- [2] BIGGS J, TANG C. Teaching for Quality Learning at University[M]. 4th ed. Buckingham: Open University Press, 2011.
- [3] China Engineering Education Accreditation Association. Engineering Education Accreditation Standards (2024 Edition)[Z]. 2024.
- [4] SPADY W G. Outcome-Based Education: Critical Issues and Answers[M]. Arlington: American Association of School Administrators, 1994.
- [5] LI Z Y. Outcome-Based Instructional Design[J]. China University Teaching, 2015(3): 32–39.
- [6] BRUNER J. Actual Minds, Possible Worlds[M]. Cambridge: Harvard University Press, 1986.
- [7] HUANG Z W. Documentary Network Novels (Seven Volumes, Teaching Excerpts S001 – S021) [M/OL]. QQ Reading/17K Literature.

- [8] LAURILLARD D. Teaching as a Design Science: Building Pedagogical Patterns for Learning and Technology[M]. New York: Routledge, 2012.
- [9] HUANG Z W. Self-compiled Lecture Notes for Graduate Course "Solid Waste Pollution Prevention and Control Technology" (2025 Edition)[Z]. Chengdu University, 2025.
- [10] PERKINS D N. Technology Meets Constructivism: Do They Make a Marriage?[J]. Educational Technology, 1991, 31(5): 18-23.
- [11] LAVE J, WENGER E. Situated Learning: Legitimate Peripheral Participation[M]. Cambridge: Cambridge University Press, 1991.
- [12] ANDERSON L W, KRATHWOHL D R. A Taxonomy for Learning, Teaching, and Assessing: A Revision of Bloom's Taxonomy[M]. New York: Longman, 2001.
- [13] WANG J. Cultivating Higher-Order Thinking Skills in Graduate Students from Bloom's Cognitive Theory Perspective[J]. Graduate Education Research, 2021(3): 45-50.
- [14] YU S Q. From Knowledge Transmission to Cognitive Construction: The Core Value of Educational Informatization[J]. Peking University Education Review, 2019, 17(1): 174-184.
- [15] ZHONG B L, SHANG J J, WANG J H, et al. ChatGPT's Challenges and Responses for Education[J]. Chongqing Higher Education Research, 2023, 11(2): 3-10.